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MANAGEMENT REVIEW

Management without theatre.



ROLE

THE MANAGER AS TRAINER: TWI JOB INSTRUCTION

How key points cut lens grinding from five years to months, a pocket card with four steps, what experts leave out when they teach, the timetable that measures training, and a job breakdown sheet.

1,005,170

Job Instruction certificates by September 1945: more than half of all those the Training Within Industry service issued.

TWI REPORT, 1945

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Most managers were never taught how to teach. They learned the job, became good at it, and were then asked to pass it on. Job Instruction, written in the Second World War and still in use, treats teaching a job as a skill of its own.

In 1940 the United States needed 350 lens grinders and thought the craft took five years to learn. Breaking the work into steps and key points cut that to months, and by 1945 more than a million Job Instruction certificates had been issued. The pocket card puts the burden on the teacher. A study of expert surgeons shows why: teaching a procedure, they left out half or more of each kind of step. TWI's timetable shows how to measure training, and the sheet on page 8 is where it starts.

Stiven Janaqi EDITOR · STIVEN CATALYST

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HOW TO READ THIS ISSUE

Figure
Every figure has its source and year at the foot of its page.

Our reading
Where the editors interpret rather than the research, it says so.

Practice
The steps and the card are proposals to try, not research results.

COVER STORY

From five years to months

In August 1940, in its first week, the Training Within Industry service (TWI) got its first task: to help US arsenals and navy yards find 350 qualified lens grinders. A learner was thought to do well to master the craft in five years.

Lens grinding turned out to be 20 jobs. New people were broken in on one simple job, and each part was studied for the few critical points that decided whether the whole operation succeeded. TWI called them key points. Training for the separate jobs fell from about five years to a matter of months.

1917–19
Charles R. Allen trains shipyard instructors in four steps: show, tell, do, check.

1940
Lens grinding: the key points are found.

1941
The method fits into ten hours; a national programme from August.

1945
1,005,170 Job Instruction certificates in all.

OUR READING

Key points are what the expert no longer notices. Once written down, they stop depending on who happens to do the teaching.

Source: War Manpower Commission, Bureau of Training, Training Within Industry Service, 1945

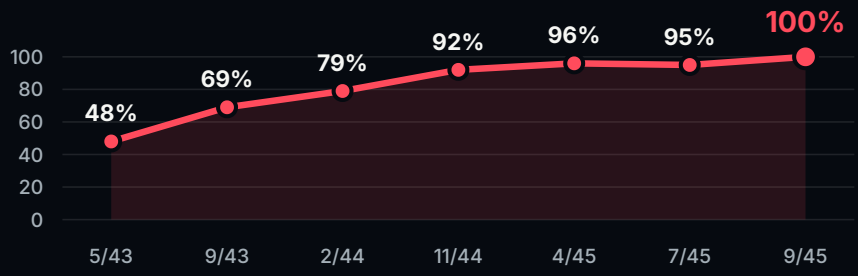
From TWI's own final report (1945). “Months” is the report's word; it gives no exact figure. The count is of certificates, and one supervisor could hold several.

THE NUMBERS

Training time, seven counts

Seven times between May 1943 and September 1945, TWI tabulated the results that plants reported. Of the plants that reported a change in training time, the share with a reduction of 25% or more:

PLANTS REPORTING TRAINING TIME CUT BY 25% OR MORE, BY TABULATION (MONTH/YEAR)



Job Instruction was the largest of TWI's programmes: 1,005,170 of the 1,750,650 certificates issued by 30 September 1945. A group of ten supervisors took it in five two-hour sessions. Some single plants reported training time cut by 90%.

OUR READING

Plants chose whether to report, and TWI used the counts to defend its budget. The curve records conviction, not a controlled test; page 6 looks for a comparison.

Source: War Manpower Commission, Bureau of Training, Training Within Industry Service, 1945

Each share is of the plants that reported a training-time result in that tabulation; the report does not say how many plants each one counted. The first used more than 600 voluntary plant statements.

THE MODEL

Get ready, then four steps

Before teaching, the card says: have a timetable, break down the job into important steps and key points (safety is always a key point), have everything ready, and arrange the workplace as the worker will be expected to keep it. Then:

1. **Prepare the worker**
   Put them at ease, find out what they already know, get them interested.
2. **Present the operation**
   Tell, show and illustrate one important step at a time; stress each key point.
3. **Try out performance**
   They do the job and explain each key point; continue until you know they know.
4. **Follow up**
   Put them on their own, name who helps, check often, taper off.

“If the worker hasn't learned, the instructor hasn't taught.”

The TWI Institute still teaches the method in 10 hours to groups of up to 10. One stated aim: to replace job shadowing and “buddy” training. According to the Lean Enterprise Institute, Job Instruction is still the primary training tool of Toyota's team leaders worldwide.

Sources: War Manpower Commission, Bureau of Training, Training Within Industry Service, 1945; Lean Enterprise Institute, 2023; TWI Institute, 2026; Lean Enterprise Institute, 2014

Card wording after the TWI report (1945) and the Lean Enterprise Institute's reprint (2023), which writes “them” where the original wrote “him”. Translations are ours.

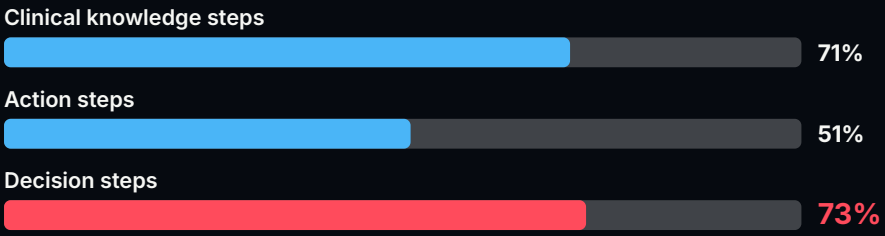
More in the essay “Watch how I do it” is not training

WHAT THE RESEARCH SAYS

What experts leave out

The 1945 report warned that a competent worker's familiarity with the work often makes him overlook the difficulties he had at the start. In 2014, Maura Sullivan and colleagues filmed three expert surgeons teaching an emergency airway procedure and compared what they said with a full task list.

STEPS LEFT OUT WHILE TEACHING, AVERAGE OF THREE EXPERTS, 2014



For action steps, the experts said how to do them only 13% of the time. Across 11,575 US firms, Nicola Bianchi and Michela Giorcelli found that Job Instruction raised sales and productivity for good, though less than Job Relations; trained firms then spent more on regular maintenance and less on machine repairs and injuries.

OUR READING

The key-point column exists for the expert's blind spot: asking what makes or breaks the job brings back what practice has made invisible.

Sources: War Manpower Commission, Bureau of Training, Training Within Industry Service, 1945; Maura E. Sullivan, Kenneth A. Yates, Kenji Inaba, Lydia Lam & Richard E. Clark, Academic Medicine, 2014 (via PubMed abstract); Nicola Bianchi & Michela Giorcelli, Journal of Political Economy, 2022

Sullivan et al.: three experts, one procedure, figures from the abstract. Bianchi and Giorcelli: a natural experiment, figures from the article (JPE, 2022).

HOW IT IS MEASURED

Until you know they know

TWI found that workers often just copied motions without understanding them. From January 1943 the manual required the learner in step 3 to give back what he was doing and why. Training is measured in three places:

01	**Timetable**	Workers down the side, jobs across the top; tick what each can already do, set a date for every gap.
02	**Give-back**	In the try-out, every key point and its reason, in the learner's own words.
03	**Follow-up**	Who they go to for help, when you check, when extra coaching ends.

HYPOTHETICAL EXAMPLE, THE TIMETABLE OF A RETURNS DESK

Team	6 people, 4 jobs: 24 boxes
Can do it	15 boxes ticked
Gaps	9: 6 with a date and a trainer, 3 with neither

The three gaps without a date are where “watch how I do it” fills in. All figures here are invented.

Source: War Manpower Commission, Bureau of Training, Training Within Industry Service, 1945

The timetable and the give-back are from the 1945 report; the example is the editors'.

TOOL OF THE ISSUE

The job breakdown sheet

One job per sheet, written where the work is done. If a KPI has turned red, first check that the cause is a matter of skill at all; the KPI diagnostic helps.

01 The job what is done, and what done well looks like

02 Important steps (what) each a segment where the work moves forward

03 Key points (how) what makes or breaks the job, could injure, or makes it easier

04 Reasons (why) why each key point matters

05 Timetable who should do it, how well, by what date

06 Try-out and follow-up key points given back; who helps; check dates

Sources: War Manpower Commission, Bureau of Training, Training Within Industry Service, 1945; Lean Enterprise Institute, 2023

Proposed by the editors, after the Job Instruction card (1945) and the three-column sheet reprinted by the Lean Enterprise Institute (2023).

Open the tool: KPI Diagnostic stivencatalyst.com/tools/kpi-diagnostic/

SOURCES AND METHOD

Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.

The Training Within Industry Report, 1940-1945
War Manpower Commission, Bureau of Training, Training Within Industry Service, 1945
https://www.govinfo.gov/content/pkg/GOVPUB-PR32_5200-00275cf1fd80a1d51995ae716fd38f34/html/GOVPUB-PR32_5200-00275cf1fd80a1d51995ae716fd38f34.htm

Lean Lexicon: Training Within Industry (TWI)
Lean Enterprise Institute, 2014
https://www.lean.org/lexicon-terms/training-within-industry-twi/

TWI training cards for printing: Job Instruction, Job Methods, Job Relations and the Job Breakdown Sheet
Lean Enterprise Institute, 2023
https://email.lean.org/hubfs/ebook%20files/TWI_training_cards_for_printing.pdf

Job Instruction (JI)
TWI Institute, 2026
https://www.twi-institute.com/job-instruction/

The Use of Cognitive Task Analysis to Reveal the Instructional Limitations of Experts in the Teaching of Procedural Skills
Maura E. Sullivan, Kenneth A. Yates, Kenji Inaba, Lydia Lam & Richard E. Clark, Academic Medicine, 2014 · via PubMed abstract
https://doi.org/10.1097/ACM.0000000000000224

The Dynamics and Spillovers of Management Interventions: Evidence from the Training Within Industry Program
Nicola Bianchi & Michela Giorcelli, Journal of Political Economy, 2022
https://www.nber.org/papers/w28833

EDITORIAL METHOD

Each figure was checked for its year, its publisher and what exactly it measures. Where the publisher's page could not be opened, the figure was checked against independent summaries and is marked “via”. The editors' interpretation is marked “Our reading”. Figures that could not be confirmed are not in the issue.



MANAGEMENT REVIEW

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Every issue, one management question, checked against the best research.

ALL ISSUES

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